

JULIEN CHAGNON

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SUMMARY

Bilingual (English-French) third year Computer Engineering student seeking a 12-16 month co-op placement to further develop industry skills and exposure. Proficient in C/C++, Python, Java, HTML/CSS/JS, and Assembly with hands-on experience in VHDL, Arduino as well as web development. Experienced in data management using SQL, spatial mapping using GeoJSON format, and CAD modeling. Adept at problem-solving, debugging, and project collaboration, with a strong foundation in computer architecture and software development.

EDUCATION

Bachelor of Applied Science in Computer Engineering – Queen's University (2023-Present)

Cumulative GPA - 3.57

- Dean's Scholar (2024/25 Academic Year) – Stephen J.R. Smith Faculty of Engineering and Applied Science, awarded for outstanding performance.
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WORK EXPERIENCE

Traffic Services Intern

City of Ottawa – May to August 2025

- Applied data analysis to collect and interpret pedestrian and vehicle survey data to improve roadway safety.
- Automated form collection workflows using Microsoft Power Automate.
- Used GIS tools to help map and analyze roadway data for traffic investigations.

Graffiti Management Assistant

City of Ottawa – May to August 2024

- Maintained a comprehensive database tracking service requests for city-wide graffiti removal efforts.
- Performed graffiti removal with industrial equipment while adhering to strict safety procedures for handling corrosive chemicals and maintaining equipment.

Camp Counsellor

Mountain Bike Kids – June to August 2022

- Supervised campers aged 8 to 14 while planning and leading mountain bike outings and day trips.
 - Provided customer service and support to parents, addressing individual needs, concerns, or complaints.
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PROJECTS

Running and Jumping Detection using machine learning (Python, scikit-learn)

- Trained logistic regression model (>95% accuracy), packaged with joblib, and built Tkinter desktop interface with real-time matplotlib visualization
- Developed end-to-end pipeline for CSV ingestion, hierarchical HDF5 storage, preprocessing, and extraction of 40+ features per 5-second window.

Road Learning Tool (React, Typescript, MapLibre GL)

- Built an interactive web application that renders custom vector tiles for performant road visualization across multiple cities and zoom levels.
 - Designed a data processing pipeline using Python to extract, normalize, and index road geometry and metadata from OpenStreetMap GeoJSON.
 - Implemented fully client-side quiz logic, filtering, and state management, enabling dynamic road highlighting.
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COMPETENCIES, SKILLS, AND ABILITIES

Programming experience: C/C++, Python, Java, Assembly (Nios II), JavaScript, VHDL

Libraries and Frameworks: Qt, Matplotlib, Pandas, Scikit-learn, React

Web Development: HTML, CSS

Operating Systems: Windows, Linux

Data Management & Visualization: SQL, Seaborn, GeoJSON format

Software and Tools: Microsoft Office, Git, LTspice, GIS tools (QGIS), and SolidWorks CAD

CERTIFICATIONS

- Ontario G Class Driver's License (Clean Record)
- Standard First Aid/CPR-C and Automated External Defibrillator Certification (2024)